1. Inheritance

1.1. Chromosomes, Genes and Proteins

- **Chromosomes:** made of DNA, which contains genetic information in the form of genes
- **Gene:** a length of DNA that codes for a protein
- **Allele:** an alternative form of a gene
- Inheritance of sex in humans is used with **X and Y chromosomes.**
- **Haploid nucleus:** a nucleus containing a single set of unpaired chromosomes (e.g., sperm and egg)
- **Diploid nucleus:** a nucleus containing two sets of chromosomes (e.g., in body cells)
- The sequence of bases in a gene determines the **sequence of amino acids used to make a specific protein.**
- Different sequences of amino acids give different shapes to protein molecules.

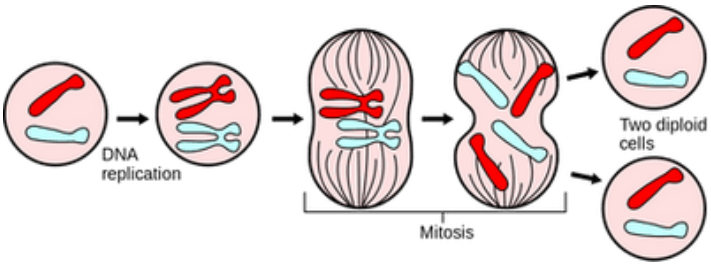
1.2. DNA and Protein Synthesis

DNA: controls cell function by controlling the production of proteins, including enzymes, membrane carriers, and receptors for neurotransmitters

- DNA has 2 long strands and 4 nucleotides, AT and CG
- mRNA has AU and CG bases
- Overall, protein synthesis occurs outside of the nucleus in the cytoplasm
- **Protein synthesis has two stages:**
  - **Transcription** (rewriting the base code of DNA into bases of mRNA)
  - **Translation** (using mRNA base sequence to build amino acids into a sequence in a protein)
- **How proteins are made:**
  - the gene coding for the protein remains in the nucleus
  - messenger RNA (mRNA) is a copy of a gene
  - mRNA molecules are made in the nucleus and move to the cytoplasm
  - the mRNA passes through ribosomes
  - the ribosome assembles amino acids into protein molecules
  - the sequence determines the specific order of amino acids of bases in the mRNA
- All body cells in an organism contain the same genes, but many genes in a particular cell are not expressed because the cell only makes the specific proteins it needs

1.4. Cell Division: Mitosis

Mitosis: The nuclear division gives rise to genetically identical cells

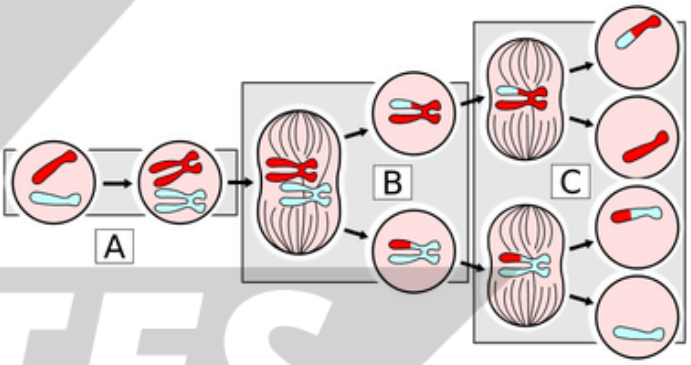


- Mitosis is needed for:
  - **Growth:** in animals, each tissue provides its own new cells when needed.
  - **Repair** damaged tissues: for example, when you cut your skin, mitosis provides new cells to cover up cuts.
  - **Replacement** of worn-out cells
  - **Asexual reproduction:** in plants
- The exact replication of chromosomes occurs before mitosis
- During mitosis, the copies of chromosomes separate, maintaining the chromosome number in each daughter cell

Stem Cells: unspecialised cells that divide by mitosis to produce daughter cells that can become specialised for specific functions

1.5. Cell Division: Meiosis

Meiosis: Reduction division in which the chromosome number is halved from diploid to haploid



- Meiosis is involved in the production of gametes.
- Meiosis results in genetic variation, so the cells produced are not all genetically identical.
- Gametes such as sperm and ovum are produced via meiosis.

1.6. Monohybrid Inheritance

Inheritance: The transmission of genetic information from generation to generation.

- Here are some common terminologies you should know to understand monohybrid inheritance.

Terminologies

- **Genotype:** the genetic makeup of an organism in terms of the alleles present (e.g. Tt or GG)

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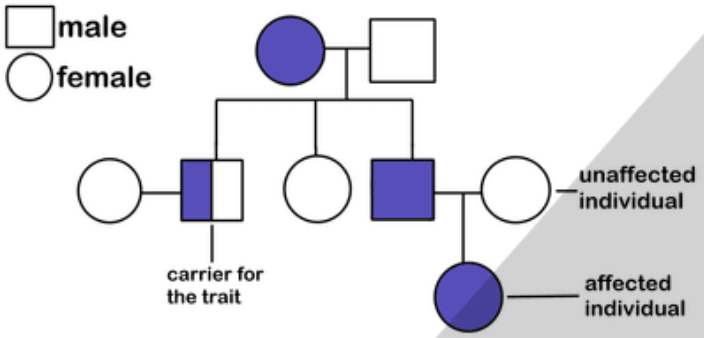
- **Phenotype:** the observable features of an organism (e.g. tall plant or green seed)

  genotype + environment + random variation → phenotype
- **Homozygous:** having two identical alleles of a particular gene (e.g. TT or gg). Two identical homozygous individuals that breed together will be pure-breeding
- **Heterozygous:** having two different alleles of a particular gene (e.g. Tt or Gg), not pure-breeding
- **Dominant:** an allele that is expressed if it is present (e.g. T or G).
- **Recessive:** an allele that is only expressed when there is no dominant allele of the gene present (e.g. t or g)

Pedigree Diagram

- Pedigree diagrams trace the **inheritance pattern** of a specific characteristic (usually a disease) **through generations of a family.**
- **Pure Breeding:** The individual is homozygous for that characteristic
- This can determine the probability that someone in the family will inherit the genetic disorder.



Genetic Diagrams

- Monohybrid Inheritance can be determined using a genetic diagram known as a **Punnett square.**
- A Punnett square diagram shows the **possible combinations of alleles** that could be produced in the offspring.
- The **dominant allele** is shown using a capital letter, and the **recessive allele** uses the same letter but lowercase.
- If you are asked to use your own letters to represent the alleles in a Punnett square, try to choose a letter that is obviously different as a capital than the lowercase so the examiner is not left doubt as to which is dominant and which is recessive.

1:1 Monohybrid Crosses

- In this cross, there is a **1:1 ratio of boy to girl**, meaning a **50%** chance of the offspring being a boy and a **50%** chance of the offspring being a girl.

A Genetic Diagram

	♀	♂
Parents' Phenotype:	Mother	Father
Parents' Genotype:	X X	X Y
Gametes:	X X	X Y

F1 Genotype:

	X	X
X	X X	X X
Y	X Y	X Y

F1 Phenotype: 2 : 2 = 1 : 1 Boy : Girl

Note the gender of the baby is determined by the sperm!

3:1 Monohybrid Crosses

- There is more variation in this cross, with a **3:1** ratio of brown eyes: blue eyes, meaning each offspring has a 75% chance of having brown eyes and a 25% chance of having blue eyes

A Genetic Diagram

	♀	♂
Parents' Phenotype:	Brown eyes	Brown eyes
Parents' Genotype:	B b	B b
Gametes:	B b	B b

F1 Genotype:

	B	b
B	B B	B b
b	B b	b b

F1 Phenotype: 3 : 1 Brown eyes : blue eyes

Note the gametes are always put in circles

Example: Inheritance of flower colour

- Let the Purple flower (P) be dominant, and the White flower (p) be recessive
- First Cross (Parent Generation): Parent 1 (PP) x Parent 2 (pp)
  - All offspring will be Pp, which are all purple flowers
  - This is the F1 generation

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- Second cross: Pp x Pp
  - Possible Allele Combinations:
    - PP (homozygous dominant)
    - Pp (heterozygous)
    - pp (homozygous recessive)
  - Phenotypes
    - 3 Purple-flowered plants (PP, Pp, Pp)
    - 1 White-flowered plant (pp)

1.7. Modification of Mendelian Genetics

Codominance

Codominance: a condition where two alleles of a gene are **equally dominant (50%)**

- Both alleles are equally expressed in the phenotype of the heterozygote.
- For example, if the parent phenotype is red and white, a co-dominant will have both red and white colour in the offspring.

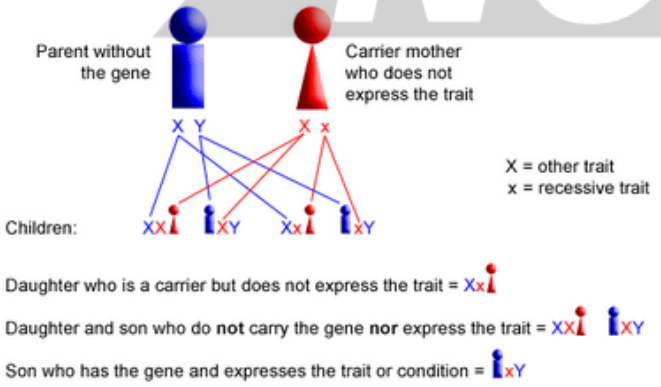
ABO Blood Group (Codominance and Multiple Allele)

- Inheritance of blood group is an example of codominance
- There are three alleles for the blood group given by the symbols I^A, I^B and I^O.
- I^A and I^B are co-dominant giving blood group AB or I^AI^B, and both dominant to I^O.

Sex-Linked Characteristics

Sex-linked characteristic: a characteristic in which the gene responsible is located on a sex chromosome, making it more common in one sex than in the other.

- Generally, in the IGCSE syllabus, the most common is **X-linked recessive trait**.
- X-linked recessive disorders are more common in males than in females.



Syllabus 17.4.18: You must be able to use genetic diagrams to predict the results of monohybrid crosses involving codominance or sex linkage and calculate phenotypic ratios.

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